Integration

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Part I

Area under the curve

Introduction

One way of describing changes in biological and physical processes across time and space is by specifying the rates of change of these processes.
Examples:

- Rate of change of population density
- Rate of change of drug concentration in the blood

Although processes may be described by their rates, we often want to know the actual values of the variables, not just their rates.
Examples:

- What is the population density?
- What is the drug concentration in the blood?

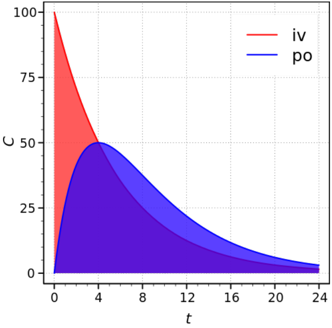
Moreover, often we are also interested in cumulative effects. E. g., the toxicity of a drug is affected by the amount of drug in the blood times the length of time it remains at that level.

Example: Bioavailability

Pharmacology: Bioavailability describes extent and rate with which an administered dose of unchanged drug reaches the systemic circulation.

Bioavailability is typically assessed by determining the "**a**rea **u**nder the plasma concentration–time **c**urve" (AUC).

Absolute bioavailability compares the bioavailability of the active drug in systemic circulation following non-intravenous administration with the bioavailability of the same drug following intravenous administration.



Absolute bioavailability is the dose-corrected ratio of areas under the curves,

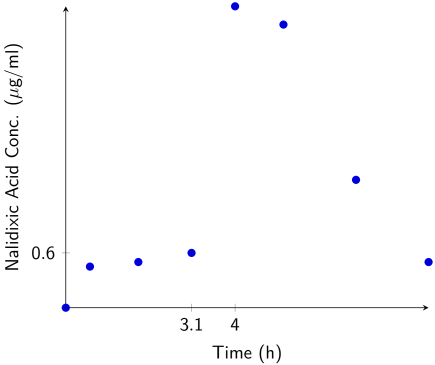
$$F_{abs} = 100 \frac{AUC_{po} \cdot D_{iv}}{AUC_{iv} \cdot D_{po}}$$

iv, intravenous; po, oral route. C is plasma concentration
Picture source: Wikipedia

Nalidixic Acid

Say we want to construct tablets of Nalidixic Acid (a synthetic antibiotic). However, doing this often results in a lower bioavailability of the active substance than administered intravenously.

To investigate this a 250 mg tablet of Nalidixic acid was administered to a patient, whereafter the concentration was measured.

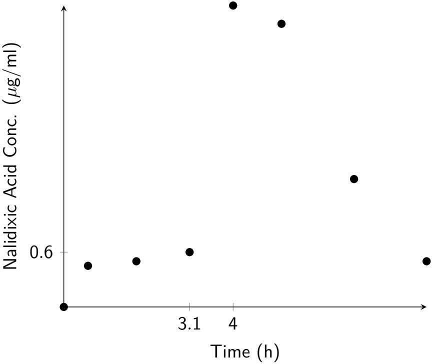


What is the area under the curve?

Ref: Murata et.al., Pharmacokinetic analysis of single- or multiple-dose plasma drug concentration data with a microcomputer using multi-fraction absorption models. J Pharm Sci. 1989

Area Under the Curve (AUC)

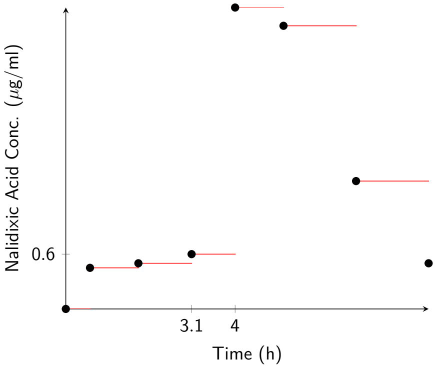
To answer this question we have to first interpolate the data and then calculate the AUC. We will present 3 ways:



Area Under the Curve (AUC)

To answer this question we have to first interpolate the data and then calculate the AUC. We will present 3 ways:

- Constant interpolation, using left end points

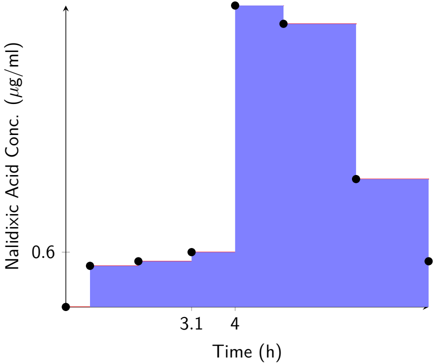


Area Under the Curve (AUC)

To answer this question we have to first interpolate the data and then calculate the AUC. We will present 3 ways:

- Constant interpolation, using left end points

$\Longrightarrow$ Left Riemann Sum
AUC $= 11.6\ \mu$g$\cdot$ h/ml



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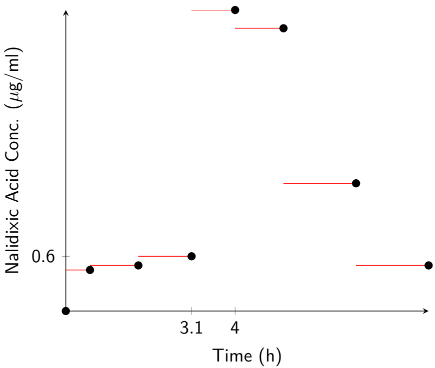
- Constant interpolation, using right end points



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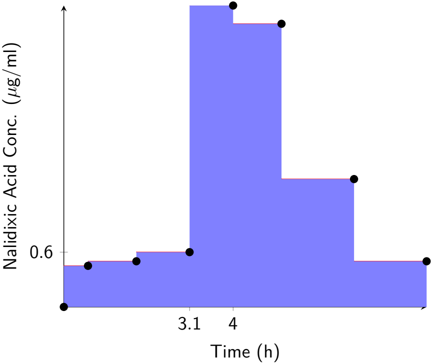
AUC $= 10.6\ \mu$g$\cdot$ h/ml



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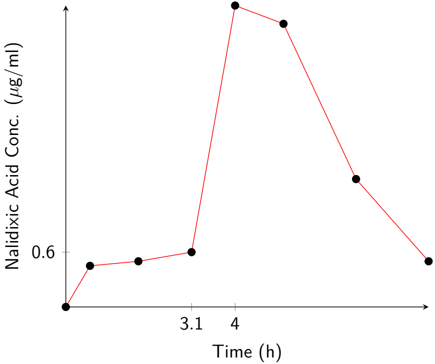
- Linear Interpolation



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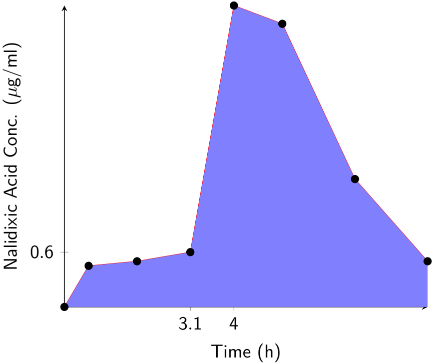
- Constant interpolation, using right end points

$\Longrightarrow$ Right Riemann Sum
AUC $= 10.6\ \mu$g$\cdot$ h/ml

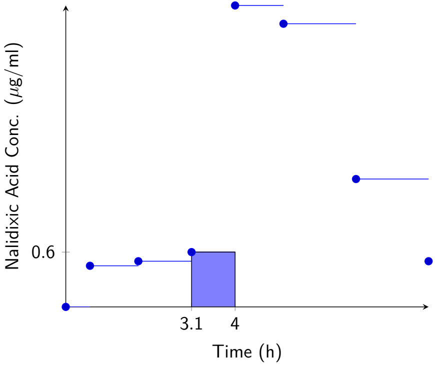
- Linear Interpolation

$\Longrightarrow$ Trapezoidal
AUC $= 11.1\ \mu$g$\cdot$ h/ml



AUC: Left Riemann

Let us first calculate the area of a single panel:

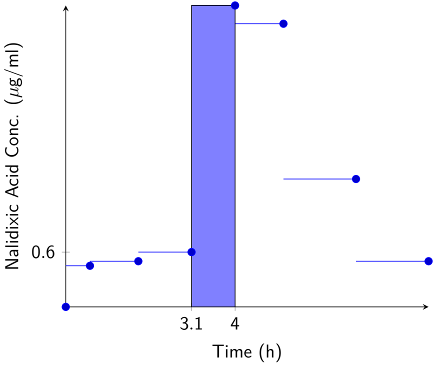


$\text{Area}_i = (x_{i+1} - x_i) * y_i$ for $i = 1, 2, \ldots, n-1$

In this case: $(4\text{h}-3.1\text{h}) * 0.6\mu\text{g/ml} = 0.54\mu\text{g}\cdot\text{h/ml}$

AUC: Right Riemann

Let us first calculate the area of a single panel:

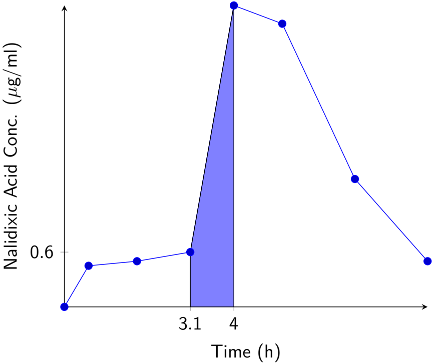


$\text{Area}_i = (x_{i+1} - x_i) * y_{i+1}$ for $i = 1, 2, \ldots, n-1$

In this case: $(4\text{h} - 3.1\text{h}) * 3.3\mu\text{g/ml} = 2.97\mu\text{g}\cdot\text{h/ml}$

AUC: Trapezoidal

Let us first calculate the area of a single panel:



$\text{Area}_i = (x_{i+1} - x_i) * \frac{y_i + y_{i+1}}{2}$ for $i = 1, 2, \ldots, n-1$

In this case: $(4\text{h} - 3.1\text{h}) * \frac{0.6+3.3}{2} \mu\text{g/ml} = 1.76 \mu\text{g}\cdot\text{h/ml}$

Composite Quadrature

he entire area is now obtained by summing all pieces together:

$$\text{Area} = \sum_{i=1}^{n-1} \text{Area}_i$$

Inserting the equation for a single panel we obtain:

- Left Riemann

$$\text{Area} = \sum_{i=1}^{n-1} (x_{i+1} - x_i) * y_i$$

- Right Riemann

$$\text{Area} = \sum_{i=1}^{n-1} (x_{i+1} - x_i) * y_{i+1}$$

- Trapezoidal

$$\text{Area} = \sum_{i=1}^{n-1} (x_{i+1} - x_i) \frac{y_i + y_{i+1}}{2}$$

NOTE: The Trapezoidal rule corresponds to the average of the left and right Riemann sum.

In R

```
time = c(0.5,1.0,2.0,3.0,4.0,5.0,6.5,8.0)
conc = c(0,0.45,0.5,0.6,3.3,3.1,1.4,0.5)

n = length(time)
lRiemann = sum((time[2:n]-time[1:n-1])*conc[1:n-1])
rRiemann = sum((time[2:n]-time[1:n-1])*conc[2:n])
trapezoidal = sum((time[2:n]-time[1:n-1])*
(conc[1:n-1] + conc[2:n])/2)
```

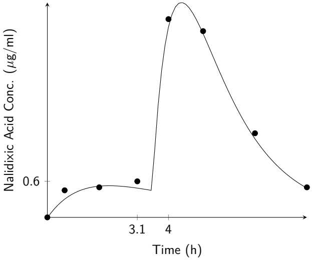
NOTE: R commands can be splitted between lines, which has been done here to fit on the slide - It is not necessary in R.

From Data to Function

After having performed the bioavailability experiment several times, the following model has been fitted to the observed data:

$$f(t) = \sum_{i=1}^{n} \alpha_i (e^{\beta_i (t - T_i)} - e^{\gamma_i (t - T_i)}) \quad \text{for } t \in [T_{n-1}, T_n] \tag{1}$$

For $n = 2$ we obtain the following fit, see ref. for parameters:



AUC for Functions

Thus we are now interested in the area under the curve given by f

This is defined to be the definite integral (Note: the outcome is a number):

$$\int_a^b f(x)\mathrm{d}x \tag{2}$$

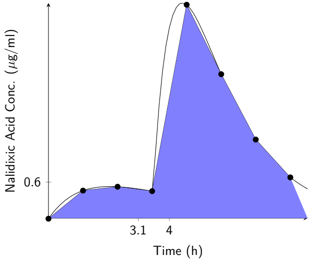
However, often it is not possible to calculate integrals explicitly

IDEA: Use the function to introduce new "datapoints" and use the techniques from before to approximate the integral!
NOW: Exact calculation of the integral

From Function to Data

We proceed as follows:

- Evaluate the function in n (usually equidistant) points,
- Ignore the old function,
- Interpolate the points (using either Left/Right Riemann or Trapezoidal),
- Calculate the area under the resulting curve.



Approximate AUC for functions

Given the integral:

$$\int_a^b f(x)\mathrm{d}x$$

Then we can approximate it with 3 different methods:

- Left Riemann

$$\int_a^b f(x)\mathrm{d}x \approx \sum_{i=1}^{n-1}(x_{i+1}-x_i)f(x_i) = \frac{b-a}{n-1}\sum_{i=1}^{n-1} f(x_i)$$

- Right Riemann

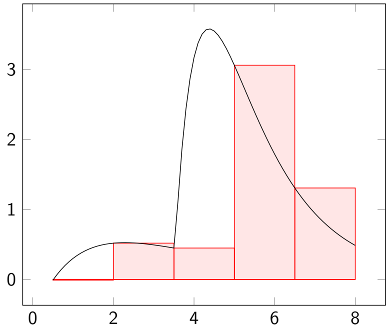
$$\int_a^b f(x)\mathrm{d}x \approx \sum_{i=1}^{n-1}(x_{i+1}-x_i)f(x_{i+1}) = \frac{b-a}{n-1}\sum_{i=1}^{n-1} f(x_{i+1})$$

- Trapezoidal

$$\int_a^b f(x)\mathrm{d}x \approx \sum_{i=1}^{n-1}(x_{i+1}-x_i)\frac{f(x_i)+f(x_{i+1})}{2} = \frac{b-a}{2(n-1)}\left(f(x_1)+f(x_n)+2\sum_{i=2}^{n-1} f(x_i)\right)$$

Precision: Left Riemann

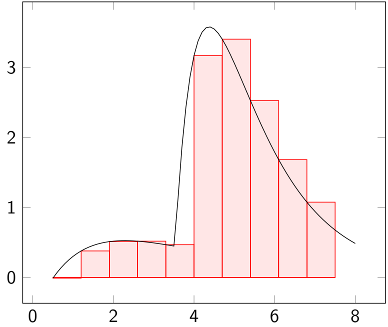
5 **Panels**



Approximate Integral: 7.9894 μg$\cdot$ h/ml
Exact Integral: 10.1118 μg$\cdot$ h/ml

Precision: Left Riemann

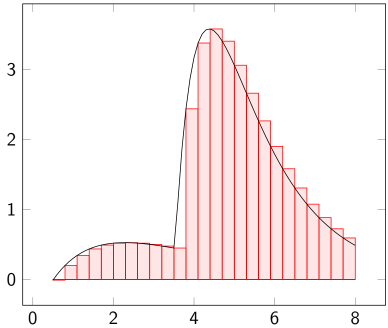
10 **Panels**



Approximate Integral: 9.4890 μg$\cdot$ h/ml
Exact Integral: 10.1118 μg$\cdot$ h/ml

Precision: Left Riemann

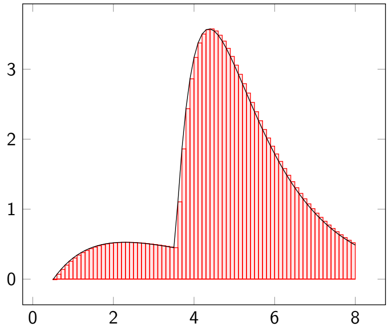
20 **Panels**



Approximate Integral: 9.9322 μg$\cdot$ h/ml
Exact Integral: 10.1118 μg$\cdot$ h/ml

Precision: Left Riemann

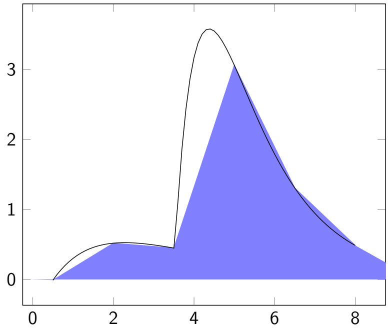
40 **Panels**



Approximate Integral: 10.0549 μg$\cdot$ h/ml
Exact Integral: 10.1118 μg$\cdot$ h/ml

Precision: Trapezoidal

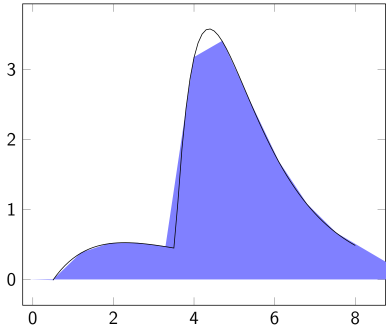
5 Panels



Approximate Integral: 8.3607 μg$\cdot$ h/ml
Exact Integral: 10.1118 μg$\cdot$ h/ml

Precision: Trapezoidal

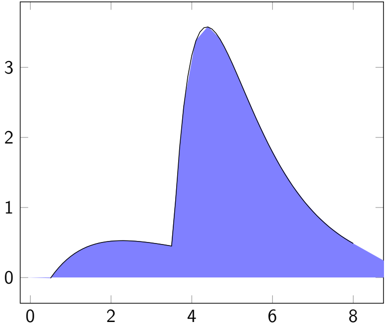
10 **Panels**



Approximate Integral: 9.6746 μg$\cdot$ h/ml
Exact Integral: 10.1118 μg$\cdot$ h/ml

Precision: Trapezoidal

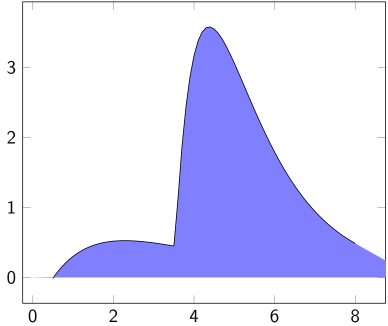
20 **Panels**



Approximate Integral: 10.0250 μg$\cdot$ h/ml
Exact Integral: 10.1118 μg$\cdot$ h/ml

Precision: Trapezoidal

40 **Panels**



Approximate Integral: 10.1013 μg$\cdot$ h/ml
Exact Integral: 10.1118 μg$\cdot$ h/ml

Which method is the best

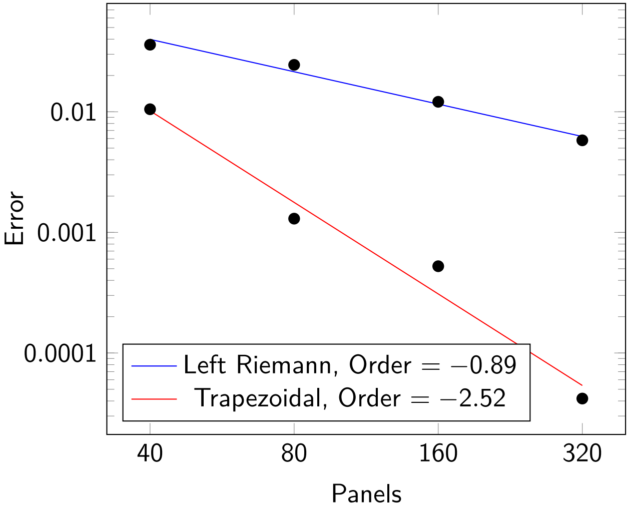
From the previous, it seems like the Trapezoidal method is the one, which gets the closer to the exact result.

To look further into this, consider the following table with the error (in μg$\cdot$ h/ml) as a function of the number of panels.

Panels	Left Riemann	Trapezoidal
40	$3.6 \cdot 10^{-2}$	$1.05 \cdot 10^{-2}$
80	$2.45 \cdot 10^{-2}$	$1.3 \cdot 10^{-3}$
160	$1.21 \cdot 10^{-2}$	$5.24 \cdot 10^{-4}$
320	$5.8 \cdot 10^{-3}$	$4.19 \cdot 10^{-5}$

Order of convergence

If we perform a linear regression on the loglog transformed data, we obtain the order (Recall: loglog corresponds to allometric model)



Part II

Introduction to integration

From Data to Function

After having performed the bioavailability experiment several times, the following model has been fitted to the observed data:

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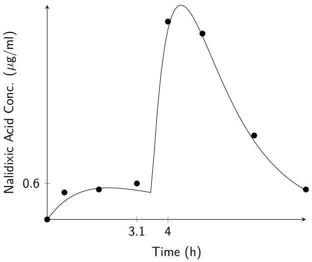
For $n = 2$ we obtain the following fit, see ref. for parameters:



AUC for Functions

Thus we are now interested in the area under the curve given by f

This is defined to be the definite integral (Note: the outcome is a number):

$$\int_a^b f(x)\mathrm{d}x \tag{2}$$

However, often it is not possible to calculate integrals explicitly

IDEA: Use the function to introduce new "datapoints" and use the techniques from before to approximate the integral!
NOW: Exact calculation of the integral

Relation between derivatives and integrals

Example (Ex. 22.1)

Consider calculating

$$\int_0^x t\mathrm{d}t,$$

i.e. finding the area under the curve $y(t) = t$ from $t = 0$ to $t = x$. This corresponds to the area of a rectangular triangle with base length x and height x, thus

$$\int_0^x t\mathrm{d}t = \frac{1}{2} \cdot x \cdot x = \frac{x^2}{2}.$$

If we define $F(x)$ such that $F(x) = \int_0^x t\mathrm{d}t = \frac{x^2}{2}$, this function $F(x)$, built from integrating $f(t) = t$, is differentiable, and

$$F'(x) = x.$$

Thus

$$\frac{\mathrm{d}}{\mathrm{d}x} \int_0^x f(t)\mathrm{d}t = f(x),$$

i.e., integration and differentiation are inverse operations.

Antiderivatives

Definition

If a function $F(x)$ satisfies $F'(x) = f(x)$, then $F(x)$ is called an *antiderivative* of $f(x)$.

Example (Ex. 22.2)

Suppose that $F'(x) = 2x$. What is $F(x)$?
We know that

$$\frac{\mathrm{d}}{\mathrm{d}x} x^2 = 2x,$$

thus x^2 is an antiderivative of $2x$.
However, also

$$\frac{\mathrm{d}}{\mathrm{d}x} [x^2 + 1] = 2x,$$

thus $x^2 + 1$ is also an antiderivative of $2x$.
More general, $x^2 + c$, where c is any constant, is an antiderivative of x.

Antiderivatives

Thus, there is not a single, unique antiderivative: If $F(x)$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x),$$

then $F(x) + c$, where c is a constant, satisfies as well

$$\frac{\mathrm{d}}{\mathrm{d}x}[F(x) + c] = f(x).$$

$F(x) + c$ is called the *family of antiderivatives* of $f(x)$.
Notation:

$$\int f(x)\mathrm{d}x = F(x) + c$$

where $\int f(x)\mathrm{d}x$ is called an *indefinite integral* and $F(z) + c$ is the family of antiderivatives of $f(x)$, where c is a constant.

Antiderivatives

Antiderivatives can often be discovered by thinking about reversing the process of taking derivatives.

Example (Ex. 22.3)

Find the family of antiderivatives of the following functions:

a) $f(x) = x^4$,

b) $f(x) = e^{3x}$.

Example (Poll everywhere)

Which is an antiderivative of $f(x) = x^2$?

Families of antiderivatives (Table 22.1)

Function $f(x)$	$F(x)$, where $F'(x) = f(x)$
0	c (any constant)
m	$mx + c$
x^n	$\frac{1}{n+1}x^{n+1} + c, \quad n \neq -1$
$\frac{1}{x}$	$\ln(\lvert x\rvert) + c, \quad x \neq 0$
$\sin x$	$-\cos x + c$
$\cos x$	$\sin x + c$
e^{ax}	$\frac{1}{a}e^{ax} + c$

Rewritten using the indefinite integral notation:

$$\int 0 \mathrm{d}x = c$$
$$\int m \mathrm{d}x = mx + c$$
$$\int x^n \mathrm{d}x = \frac{1}{n+1}x^{n+1} + c, \quad n \neq -1$$
$$\int \frac{1}{x} \mathrm{d}x = \ln(|x|) + c, \quad x \neq 0$$
$$\int \sin x \mathrm{d}x = -\cos x + c$$
$$\int \cos x \mathrm{d}x = \sin x + c$$
$$\int e^{ax} \mathrm{d}x = \frac{1}{a}e^{ax} + c$$

Properties of indefinite integrals

Given functions $f(x)$ and $g(x)$ and constants $a < b < c$ and k, it holds

(1) $\int kf(x)\mathrm{d}x = k\int f(x)\mathrm{d}x$,

(2) $\int(f(x)+g(x))\mathrm{d}x = \int f(x)\mathrm{d}x + \int g(x)\mathrm{d}x$,

Example

Find $\int(x-1)\mathrm{d}x$.

Fundamental Theorem of Calculus

If f is continuous on the interval $[a, b]$, then the function F defined by

$$F(x) = \int_a^x f(t)\mathrm{d}t \text{ for } a \leq x \leq b$$

is continuous on $[a, b]$ and differentiable on (a, b), and

$$F'(x) = f(x),$$

i.e., F is an antiderivative of f.
Moreover, if $F(z)$ is any antiderivative of f, then

$$\int_a^b f(x)\mathrm{d}x = F(b) - F(a).$$

We will also use the notation $F(b) - F(a) = F(x)|_a^b$.

Example (Ex. 22.5)

Find the derivative of the function $g(x) = \int_0^x \sin(1 + t^2)\mathrm{d}t$.

Fundamental Theorem of Calculus

Example (Ex. 22.9)

Find $\int_0^3 (x-1)\mathrm{d}x$.

Properties of integrals

Given functions $f(x)$ and $g(x)$ and constants $a < b < c$ and k, it holds

(1) $\int_a^b kf(x)\mathrm{d}x = k\int_a^b f(x)\mathrm{d}x$,

(2) $\int_a^b (f(x) + g(x))\mathrm{d}x = \int_a^b f(x)\mathrm{d}x + \int_a^b g(x)\mathrm{d}x$,

(3) $\int_a^b f(x)\mathrm{d}x + \int_b^c f(x)\mathrm{d}x = \int_a^c f(x)\mathrm{d}x$,

(4) $\int_a^b f(x)\mathrm{d}x = -\int_b^a f(x)\mathrm{d}x$.

Average values

The *average value of a function* $f(x)$ over the interval $[a, b]$ is given by

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) \mathrm{d}x.$$

Example (Ex. 22.11, Tendon strain)

The human gastrocnemius tendon is located in the calf. The force on the gastrocnemius tendon as it is stretched can be approximated by the function

$$f(x) = 71.3x - 4.15x^2 + 0.434x^3,$$

where x is the tendon elongation in mm and $f(x)$ is the force exerted by the tendon measured in Newton (N). What is the average force on the tendon as it is elongated from 2 mm to 11 mm?